



Maritime &  
Coastguard  
Agency

Guidance

# **MGN 649 (M) Amendment 1**

## **Application of MARPOL Annex II to offshore support vessels**

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## Summary

This Marine Guidance Note (“MGN”) sets out what the amendments to the International Convention for the Prevention of Pollution from Ships (“MARPOL”) Annex II and the International Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk (“IBC Code”) which came into force on 1 January 2021 mean for existing offshore support vessels (“OSVs”).

This MGN has now been amended because the transitional period outlined in the original MGN 649 (M) has come to a close and amendments to the Merchant Shipping (Prevention of Pollution from Noxious Liquid Substances in Bulk) Regulations 2018 which implement the OSV Chemical Code in the UK have now come into force. This MGN also provides some further technical guidance.

## 1. Introduction

1.1 This MGN explains how amendments to MARPOL Annex II (control of pollution by noxious liquid substances in bulk) and the IBC Code, which came into force on 1 January 2021, impact on vessels operating under the Guidelines for the Transport and Handling of Limited Amounts of Hazardous and Noxious Liquid Substances in Bulk on Offshore Support Vessels (“OSV Guidelines”) and the Code for the Transport and Handling

of Hazardous and Noxious Liquid Substances in Bulk on Offshore Support Vessels (“OSV Chemical Code”), and provides guidance to owners and operators of existing OSVs, certified in accordance with the OSV Guidelines, on how they can comply with these amended requirements.

1.2 The OSV Guidelines were adopted by the International Maritime Organization (“IMO”) under Resolution A.673(16) (the OSV Guidelines are referred to in this MGN as “A.673(16)”) and the OSV Chemical Code was adopted by the IMO under Resolution A.1122(30) (the OSV Chemical Code is referred to in this MGN as “A.1122(30)”). IMO Resolution A.1122(30) superseded, but did not revoke, Resolution A.673(16). Neither Resolution contains a mandatory requirement under MARPOL; they are guidelines referred to in regulation 11.2 of Annex II (see 2.2 below).

1.3 As this MGN can only provide a general guide and the effects of the recent amendments will vary from ship to ship, owners/operators should discuss their individual needs with their Class Society.

1.4 Marine Information Note (“MIN”) 651 (M) provided further information about the detail of the amendments to MARPOL Annex II and the IBC Code which will come into force on 1st January 2021.

## **2. Background**

2.1 All ships carrying hazardous and noxious liquid chemicals in bulk are subject to the Merchant Shipping (Prevention of Pollution from Noxious Liquid Substances in Bulk) Regulations 2018 (S.I. 2018/68) (“NLS Regs 2018”). These Regulations give effect to Annex II of MARPOL.

2.2 Regulation 11.2 of Annex II requires parties to establish appropriate measures in respect of ships other than chemical tankers and liquefied gas carriers which are certified to carry the noxious liquid substances in bulk set out in chapter 17 of the IBC Code, based on IMO guidelines. These guidelines include alternatives to the standards in the IBC Code for OSVs which carry certain hazardous and noxious liquid substances in bulk, and so OSVs will accordingly satisfy the carriage requirements under Chapter VII of the Annex to the International Convention for the Safety of Life at Sea, 1974 (SOLAS) and Annex II to MARPOL, provided that they meet the equivalent requirements of:

1. for OSVs, the keels of which were laid or which were at a similar stage of construction, on or after 19 April 1990 and before 1 July 2018, the OSV Guidelines, as amended;

2. for OSVs, the keels of which were laid or which were at a similar stage of construction, on or after 1 July 2018, the OSV Chemical Code.

2.3 Amendments to the IBC Code in IMO Resolution MEPC.318(74), came into force both internationally and in the UK on 1 January 2021 and imposed revised, and in many cases, stricter carriage requirements for products listed in Chapter 17 of the IBC Code and the latest edition of the MEPC.2/Circular (provisional categorization of liquid substances in accordance with MARPOL Annex II). These revised carriage requirements augmented existing measures to prevent pollution of the marine environment with measures aimed at safeguarding life, health and material values. Consequently, many products previously assigned only with a pollution hazard (P), have now also been assigned with a safety hazard (S), and in some cases have also been categorised as toxic products. Operators should also be aware that some product names have changed.

2.4 These amendments to the IBC Code resulted in many of the products listed in Appendix 1 to A.673(16) (Table of Permitted Cargoes) having upgraded carriage requirements. Consequently, existing OSVs complying with the requirements of A.673(16) would only have been able to load and carry a more restricted range of products based on the following criteria:

1. products which for safety reasons may be assigned for carriage on a type 3 ship as defined in the IBC Code and which are not required to meet the requirements for toxic products in section 15.12 of that Code,
2. noxious liquid substances which would be permitted for carriage on a type 3 ship;
3. flammable liquids.

2.5 For OSVs intending to carry a broader, less restricted range of the products listed in Chapter 17 of the IBC Code or the latest edition of the MEPC.2/Circular from 1 January 2021 onwards, full compliance with the IBC Code, or alternatively full compliance with the OSV Chemical Code, would have been required. As such, further explanation and guidance for the carriage of products on existing OSVs has been developed and is set out below and in Annex I of this document.

2.6 The recent amendments to the IBC Code also introduced product entries in Chapter 17 for offshore contaminated bulk liquids. Further explanation and guidance for existing OSVs backloading contaminated bulk liquid is set out below and in Annex II of this document.

2.7 It is reiterated that, regardless of their date of build, for all OSVs the discharge of tank washings or residues containing noxious liquid substances (NLS) or residues, or mixtures containing such substances,

must be to an offshore installation or to a shore reception facility. This is explained further in section 5 of this MGN below.

### **3. Post-Transitional Period Guidance**

3.1 The United Kingdom recognised that the coming into force of the amendments to MARPOL Annex II and the IBC Code on 1 January 2021 posed a potentially complex challenge for existing A.673(16) OSVs to comply with the revised carriage requirements in Chapter 17 of the IBC Code. The United Kingdom also recognized the potential difficulties for existing OSVs to comply with the higher standard in A.1122(30).

3.2 Accordingly, the United Kingdom allowed existing OSVs complying with A.673(16) to continue to operate in UK waters, subject to their Certificate of Fitness (“CoF”) remaining valid and carrying only products listed in Appendix 1 of A.673(16), until 31 December 2021. This transitional period was a delay to the implementation of the recent IBC Code amendments, not a permanent exemption, in order to give owners and operators of existing OSVs some additional time to consider either transitioning to the requirements of A.1122(30) in order to carry the broader range and quantity of chemical products as permitted under A.1122(30), or continuing to operate under A.673(16) and carrying only those products permitted under A.673(16).

3.3 Now that this transitional period has ended, existing OSVs that have chosen to continue to operate as A.673(16) vessels should follow the product carriage requirements in Annex I of this MGN.

3.4 Existing OSVs that have been complying with A.673(16) for some time will be familiar with MGN 283 – Guidance on the Back Loading of Contaminated Bulk Liquids from Offshore Installations to Offshore Supply/Support Vessels. Additional information regarding the carriage of contaminated bulk liquids and the generation of Hydrogen Sulphide (H<sub>2</sub>S) is now contained in Chapter 16 of A.1122(30), which has been developed from, and builds on, MGN 283. Now that this transitional period has ended, the requirements of Chapter 16 of A.1122(30) shall be followed by all OSVs (A.673(16) and A.1122(30)) carrying offshore contaminated bulk liquids. See Annex II to this MGN below for further information on the carriage of contaminated bulk liquids.

3.5 The requirements in this MGN apply to OSVs operating in UK waters. If the vessel is to operate in other waters outside the UK, then the

owner/operator of the vessel is advised to contact the relevant Administration to discuss its requirements accordingly.

3.6 Owners/operators of OSVs should discuss the specific requirements for their vessels with their Class Society in the first instance.

## **4. Technical Guidance**

4.1 The NLS Regs 2018, as the regulations which give effect to Annex II of MARPOL in the UK, have been amended to include both A.673(16) and A.1122(20) as recognised certification standards for OSVs carrying noxious liquid substances in bulk. However, to date, there have been very few if any vessels newly built or converted to the requirements of A.1122(20). Therefore, currently there are very few illustrative examples of vessel conversions from which guidance on good practice may be shared. However, the MCA is aware of some conversions due to take place within the next year and so will look to publish further guidance on this specific subject in due course.

4.2 The MCA has previously published MGN 282 on the carriage of dangerous goods on OSVs. This guidance includes a section on the use of portable tanks on deck as part of a deck spread. This guidance has now been incorporated into A.1122(20) as chapter 17. For the sake of good order and avoidance of doubt, the MCA expects all OSVs to continue to follow MGN 282. In the case of OSVs certified under A.1122(20), the MCA's view is that compliance with the requirements of MGN 282 for deck spreads on OSVs would also satisfy the requirements of chapter 17 of A.1122(20).

4.3 The MCA is aware that some jurisdictions outside of the UK also permitted similar transitional periods to that in operation in the UK between 1 January 2021 and 31 December 2021. Such transitional periods in other jurisdictions may also have ended or may remain only partially in place. All OSV owners and operators should ensure that, when operating outside of the UK, they comply with the requirements of the jurisdiction in which they are operating. The transitional period outlined in MGN 649 (M) for UK flagged OSVs or OSVs operating in UK waters ended on 31 December 2021.

4.4 Although the transitional period has ended, where existing OSVs cannot comply with requirements brought about by the amendments to MARPOL Annex II and the IBC Code outlined above, alternative arrangements may, without guarantee, be considered. Owners/operators of OSVs should discuss this with the vessel's Class Society in the first

instance before contacting the MCA through their local Marine Office. In the case of other OSVs operating in UK waters this would be through the consideration of proposals made to and agreed with the relevant Flag State Administrations and their ROs to which the MCA may or may not wish to raise objections or add conditions for operations in UK waters. Such proposals may typically include recommendations from a Class Society following Hazard Identification (“HAZID”) analysis, identify the specific products intended for carriage in limited quantities, state a specific route or operating location within UK waters etc.

## **5. Basis of prohibition on discharges**

5.1 Many existing OSVs currently operating around the UK are certified under A.673(16) which was first developed through the IMO in 1990 and last amended in 2006.

5.2 Section 4.2 of A.673(16) states:

Discharge into the sea of residues of Noxious Liquid Substances permitted for carriage in Ship Type 3, or products listed in appendix 1 or ballast water, tank washings, or other residues or mixtures containing such substances, is prohibited. Any discharges of residues and mixtures containing noxious liquid substances should be to reception facilities in port. As a consequence of this prohibition, the Administration may waive the requirements for efficient stripping and underwater discharge arrangements in MARPOL 73/78, Annex II.

5.3 Consequently, using powers under the former regulations implementing MARPOL Annex II in the UK (i.e. Regulation 27 of the Merchant Shipping (Control of Pollution by Noxious Liquid Substances in Bulk) Regulations 1987, and Regulation 1(6) of the Merchant Shipping (Dangerous Goods and Marine Pollutants) Regulations 1990), the MCA issued MSN 1458 to exempt OSVs carrying limited amounts of hazardous and noxious liquid substances in bulk from some limited parts of those regulations on the basis that such OSVs were to comply instead with the prohibition under section 4.2 of A.673(16).

5.4 A similar prohibition on discharge is contained in the OSV Chemical Code, but rather than allowing a waiver the equivalent section, 12.2 of A.1122(20), states:

Discharge into the sea of residues of noxious liquid substances permitted for carriage under the present Code, tank washings, or other residues or mixtures containing such substances, is prohibited. Any discharges of

residues and mixtures containing noxious liquid substances should be to port reception facilities. As a consequence of this prohibition, there are no requirements for efficient stripping and underwater discharge arrangements in MARPOL Annex II.

5.5 In August 2021 the NLS Regs 2018 were amended in order to implement the OSV Chemical Code as another recognised certification standard for OSVs carrying noxious liquid substances in bulk and also to clarify the language of the NLS Regs 2018 which reiterates that OSVs are not permitted to discharge such substances or washings into the sea (see Regulation 24: <https://www.legislation.gov.uk/ukxi/2018/68/contents> (<https://www.legislation.gov.uk/ukxi/2018/68/contents>)).

5.6 For the sake of good order, the MCA requires all OSVs to continue to discharge their residues from the carriage of noxious liquid substances in bulk to reception facilities and expects such operations to be reflected in the relevant MARPOL Annex II related certification and documentation issued to the vessel.

## **6. Requirements for the training of personnel**

6.1 Paragraph 15.3 of A.1122(30) requires that all personnel should be adequately trained in the use of protective equipment and have basic training in the procedures appropriate to their duties necessary under emergency conditions. In addition, personnel involved in NLS bulk cargo operations should be adequately trained in handling procedures commensurate with their responsibilities.

6.2 Officers should be trained in emergency procedures to deal with conditions of leakage, spillage or fire involving the cargo and a sufficient number of them should be instructed and trained in essential first aid for cargoes carried, based on the guidelines developed by the Organization.

6.3 Therefore, it should be expected that personnel on existing OSVs will be subject to the same training requirements as personnel on board an OSV certified to comply with the OSV Chemical Code.

6.4 However, presently there are no specific provisions adopted under the auspices of the IMO for such training for the crew on an OSV. The most relevant training requirements are those which apply in respect of chemical tankers. Topics covered in chemical tanker training courses may typically include, but are not limited to: ship types and stability; certificates



of fitness; the IBC code; physical and chemical properties; material safety data sheets; volatility and flammability; sources of ignition; static electricity and earthing; reactivity and stowage compatibility; tank coatings; gauging and venting systems; gas detection equipment; inert gas systems; temperature control; pumps and hoses; pressure surges; tank cleaning; firefighting and personal protective equipment.

6.5 Therefore, it is recommended that owners/operators of existing A.673(16) and A.1122(30) OSVs consider appropriate guidance and training for their crews so they fully understand the hazards when loading and carrying products listed in Annex I of this MGN and Chapter 17 and 18 of the IBC Code, and the MEPC.2/Circ.

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## **Annex I - Carriage of products listed in Appendix I of A.673(16) on existing OSVs following amendment**

# MEPC.318(74) of the IBC Code (in force since 1st January 2021)

## (1) Products with no substantial changes in carriage requirements

- Oil based mud containing mixtures of products listed in Chapter 17 and 18 of the IBC Code and permitted to be carried under paragraph 1.2 of these Guidelines
- Water based mud containing mixtures of products listed in Chapter 17 and 18 of the IBC Code and permitted to be carried under paragraph 1.2 of these Guidelines
- Acetic acid
- Ethyl Alcohol
- Triethylene Glycol
- Potassium Chloride Solution
- Potassium Chloride Solutions (less than 26%)
- Sodium chloride solution (less than 30%)
- Noxious liquid, NF, (7) n.o.s. (trade name ....., contains ....) ST3, Cat. Y
- Noxious liquid, F, (8) n.o.s. (trade name ....., contains ....) ST3, Cat. Y
- Noxious liquid, NF, (9) n.o.s. (trade name ....., contains ....) ST3, Cat. Z
- Noxious liquid, F, (10) n.o.s. (trade name .., contains ....) ST3, Cat. Z
- Noxious liquid, (11) n.o.s. (trade name ....., contains .....) Cat. Z
- Non-noxious liquid, (12) n.o.s. (trade name ....., contains ....) Cat. OS
- Liquid carbon dioxide
- Liquid nitrogen

## (2) Selected offshore related brine, mud and glycol

The following products are assigned with a safety hazard (S). However, experience indicates that the cargo tanks and connected cargo transfer system which are subject to the waiver for “pollution hazard only substances having a flash point exceeding 60°C” in A.673(16) i.e. 3.1.10, 3.2.4 and 3.4.6 may be considered as adequate for these substances. Therefore, these products may continue to be carried in such cargo tanks.

- Drilling brines (containing calcium bromide)
- Calcium chloride solution (less than 35%)
- Calcium nitrate/Magnesium nitrate/Potassium chloride solution

- Calcium Nitrate Solution (50% or less)
- Cesium formate solution
- Potassium Formate Solutions
- Ethylene Glycol

(3) Please note that a suitably marked decontamination shower and eyewash shall be available on deck in a convenient location, as required in paragraph 5.1 in Resolution A.673(16) (and paragraph 14.4.2 in A.1122(30)).

(4) Ship type 2 products and toxic

Existing OSVs having a CoF issued under the provisions of A.673(16), may be certified to carry Ship type 2 products and the toxic products which are currently listed in Appendix I of A.673(16), based on the transitional arrangements above.

(5) The vessel shall comply with the summary of minimum requirements for the product as listed in chapter 17 of the IBC Code (as amended), with the following modifications:

a. the requirements for ship survival capability and location of cargo tanks for a type 2 ship in the IBC Code, can be replaced by Chapter 2 - Stability and cargo tank location in A.673(16),

b. inerting of cargo tanks which is carrying products with a flashpoint not exceeding 60°C is required, and

c. the requirements described in 15.12.2 in the IBC Code regarding a connection for a vapour-return line to shore may not be fulfilled.

(6) This is applicable for the following Ship type 2 products:

- Drilling brine (containing zinc chloride)
- Sulphuric acid
- Xylene

and the following toxic products:

- Ethylene Glycol Monoalkyl Ether
- Formic Acid (85% or less acid)
- Formic Acid (over 85%)
- Hydrochloric acid

- Methyl Alcohol
- Sodium Silicate Solution
- Sulphuric acid [which is also Ship Type 2]
- Toluene

(7) And the following contaminated backload bulk liquids (which do not appear in Appendix 1 of A.673(16), however they are added here to be consistent with Annex II of this MGN, the recent amendments to the IBC Code and current OSV operational practices)

- Offshore contaminated bulk Liquid P (o)
- Offshore contaminated bulk Liquid Treated (containing less than 0.8% of an H<sub>2</sub>S Scavenger) (o)

#### (8) Cargo tank vent systems

For OSVs not complying with A.1122(30), either after the renewal of their CoF or after 1 January 2022, but continuing to comply with A.673(16), it should be noted that products listed in this Annex which require controlled venting arrangements may continue to be carried on such vessels subject to them complying with paragraph 3.4.4 and 3.6.2 of A.673(16) for integral or independent tanks with the set point of the pressure side of the P/V valves set at a minimum 0.6 bar gauge (consistent with paragraph 4.2.7 of A.1122(30)). For existing A.673(16) vessels that cannot comply with this, alternative arrangements may be considered and should be discussed with the vessel's Class Society in the first instance.

#### (9) Cargo tank gauging systems

For OSVs not complying with A.1122(30), either after the renewal of their CoF or after 1 January 2022, but continuing to comply with A.673(16), consideration should be given to the carriage of products listed in this Annex which require restricted or closed gauging arrangements and whether modifications to current cargo tank gauging systems are required. It should also be noted that existing A.673(16) vessels might have alternative arrangements for high-level alarms agreed under paragraph 3.13 of A.673(16) for which cargo tank gauging systems were taken into consideration. Individual arrangements should be considered and discussed with the vessel's Class Society.

# **Annex II - Carriage of offshore contaminated bulk liquid on existing OSVs having a certificate issued under the provisions of the A.673(16) following the amendment of the IBC Code (coming into force on 1st January 2021)**

(1) The requirements of A.1122(30) Chapter 16 shall be observed by all OSVs carrying contaminated bulk liquids.

(2) OSVs having a CoF issued under the provisions of the A.673(16) may be certified to carry offshore contaminated bulk liquid P (o) and Offshore contaminated bulk Liquid Treated (containing less than 0.8% of an H<sub>2</sub>S Scavenger) (o) only, provided that they fulfil the requirements for relevant entry, indicated below.

(3) OSVs having a CoF issued complying with the provisions of the A.1122(30) Code, may be certified to carry Offshore contaminated bulk Liquid P (o), Offshore contaminated bulk Liquid Treated (containing less than 0.8% of an H<sub>2</sub>S Scavenger) (o) and Offshore contaminated bulk Liquid S (o).

(4) A.673(16) vessels should comply with the summary of minimum requirements for the product as listed in chapter 17 of the IBC Code. However, the requirements for ship survival capability and location of cargo tanks for a Ship Type 2 in the IBC Code, may be replaced by Chapter 2 - Stability and cargo tank location in A.673(16).

(5) The Master of the OSV should not accept loading of any contaminated bulk liquid which is not properly documented in accordance with Chapter 16.3 of A.1122(30).

(6) “Offshore contaminated bulk Liquid P (o)”

The shipper and/or the owner of the cargo should ensure the master of the OSV that this entry can be used by confirming that the following are fulfilled:

- is pollutant only and will not present any safety hazards or where the pre-backloading tests do not indicate any safety hazards (the backload

may contain components with safety hazards, as long as they are so diluted that the final mixture presents no safety hazard);

- has a flashpoint greater than 60°C; or
- will not have the potential to become more hazardous during transport.

This shall be confirmed by the Analysis form and the conclusions of the test result in paragraph 16.3.2.15 of A.1122(30).

Offshore contaminated bulk Liquid P (o)

|          |          |          |          |          |           |             |          |          |          |          |          |
|----------|----------|----------|----------|----------|-----------|-------------|----------|----------|----------|----------|----------|
|          |          | <b>f</b> | <b>g</b> | <b>h</b> |           | <b>i'''</b> | <b>j</b> | <b>k</b> | <b>l</b> | <b>n</b> | <b>o</b> |
| <b>d</b> | <b>e</b> |          |          |          | <b>i'</b> | <b>i''</b>  |          |          |          |          |          |

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|          |          |             |           |           |          |          |            |          |           |           |           |                |
|----------|----------|-------------|-----------|-----------|----------|----------|------------|----------|-----------|-----------|-----------|----------------|
| <b>P</b> | <b>2</b> | <b>Open</b> | <b>2G</b> | <b>No</b> | <b>-</b> | <b>-</b> | <b>Yes</b> | <b>O</b> | <b>No</b> | <b>AC</b> | <b>No</b> | <b>15.19.6</b> |
|----------|----------|-------------|-----------|-----------|----------|----------|------------|----------|-----------|-----------|-----------|----------------|

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(7) “Offshore contaminated bulk Liquid Treated (containing less than 0.8% of an H<sub>2</sub>S Scavenger) (o)” The shipper and/or the owner of the cargo should ensure the master that this entry can be used by confirming that the following are fulfilled:

- has been treated to remove or prevent breakout of H<sub>2</sub>S;
- is considered as pollutant only and does not present any safety hazards or where the pre-backloading tests do not indicate any safety hazards (the backload may contain components with safety hazards, as long as they are so diluted that the final mixture presents no safety hazard);
- has a flashpoint greater than 60°C; or
- will not have the potential to become more hazardous during transport.

This shall be confirmed by the Analysis form and the conclusions of the test result in paragraph 16.3.2.15 of A.1122(30).

Offshore contaminated bulk Liquid Treated (containing less than 0.8% of an H<sub>2</sub>S Scavenger) (o)

| d | e | f    | g  | h  | i' | i'' | i''' | j | k  | l  | n  | o                 |
|---|---|------|----|----|----|-----|------|---|----|----|----|-------------------|
| P | 2 | Cont | 2G | No | -  | -   | Yes  | C | No | AC | No | 15.15,<br>15.19.6 |

However, it should be noted that the treatment may not be 100% effective in controlling the formation of H<sub>2</sub>S for a long period.

The master of the OSV should not accept any back loads of any contaminated bulk liquid which is not properly documented in accordance with MGN 283(M) of 16.3 of A.1122(30) i.e. paragraph 16.4.1.1.

(8) When loading and carrying any Offshore contaminated bulk Liquid it is recommended that H<sub>2</sub>S and LEL detection equipment is available onboard the vessel either fixed or portable to monitor the atmosphere in the tank, in addition personnel working on deck should be provided with portable H<sub>2</sub>S monitoring equipment.

H<sub>2</sub>S precautions:

- Contaminated bulk liquid should be discharged from the vessel as soon as possible, preferably at the first port of call.
- The need to clean the dirty tanks should be reviewed on each voyage to minimize the risk of biological activity and H<sub>2</sub>S build up from any residue.
- Prior to backloading to a dirty tank, the potential for biological activity resulting in H<sub>2</sub>S in the dead volume and sludge should be considered. The offshore analysis of the previous contaminated bulk liquid should be compared with analyses of a sample representative for the liquid when unloading.
- If H<sub>2</sub>S or flammable vapour is detected during loading of contaminated bulk liquids the transfer should be stopped immediately.
- Vessel-specific procedures for measures to be taken when H<sub>2</sub>S is detected during loading, transport, discharge and cleaning of contaminated bulk liquids should be included in the vessel's safety management system.

(9) Additional information on the hazards of H<sub>2</sub>S can be found in A.1122(30) Chapter 16.



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